

BUSINESS INTELLIGENCE | DATA ANALYTICS REPORT

YouTube Trending Video Analytics

End-to-End Data Analytics, Statistical Testing & Business Intelligence Project

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Tools: Python | Pandas | NumPy | Matplotlib | Seaborn | Plotly | SciPy | Jupyter Notebook

Executive Summary

This report presents an end-to-end analysis of a YouTube Trending Videos dataset covering 199 trending videos across 7 content categories and 199 unique channels, published mainly across July-August. The analysis combines data cleaning, feature engineering, exploratory data analysis, statistical hypothesis testing, and business-focused recommendations to help content and marketing teams understand what drives a video to trend, and how to optimize content strategy, upload timing, and engagement.

Key Performance Indicators

199 Total Trending Videos	97.89M Total Views	4.01M Total Likes	290.6K Total Comments
491.9K Average Views / Video	4.19% Avg Engagement Rate	7 Content Categories	199 Unique Channels

68% of trending videos were published on a weekend, and **100%** were uploaded in HD, indicating HD is now a baseline expectation rather than a differentiator on the trending shelf.

1. Dataset Overview

The dataset contains 51 columns describing each trending video's metadata, engagement metrics, publish-time attributes, and pre-engineered analytical features. It combines raw YouTube API fields (title, description, tags, view/like/comment counts, publish timestamp) with engineered business features created during preprocessing (engagement rate, duration category, view bucket, upload session, etc.).

Metric	Value
Total Rows (Videos)	199
Total Columns	51
Unique Categories	7
Unique Channels	199
Date Range	July – August (Trending Window)
Missing Values (post-clean)	0
Duplicate Rows	0

Key Columns & Business Meaning

- view_count / like_count / comment_count** — Core popularity & engagement signals used across every business question.
- engagement_rate / like_rate / comment_rate** — Engineered ratios that normalize engagement by views — enable fair comparison between small and viral channels.
- duration_seconds / duration_category** — Video length in seconds, bucketed into Short/Medium/Long — used to test the 'ideal video length' hypothesis.
- publish_session / publish_hour / is_weekend** — Upload-timing features used for upload-time optimization analysis.
- category_name / category_id** — YouTube content category — the primary segmentation dimension for content strategy.
- title_length / title_word_count / tag_count** — Content-metadata features tested for correlation with reach and discoverability.
- is_hd / has_caption** — Production-quality flags used to assess HD/caption strategy impact on performance.
- view_bucket** — Quantile-based bucket (Low/Medium/High/Very High) used for cohort-style comparison.

2. Data Cleaning & Feature Engineering

Data Cleaning Steps

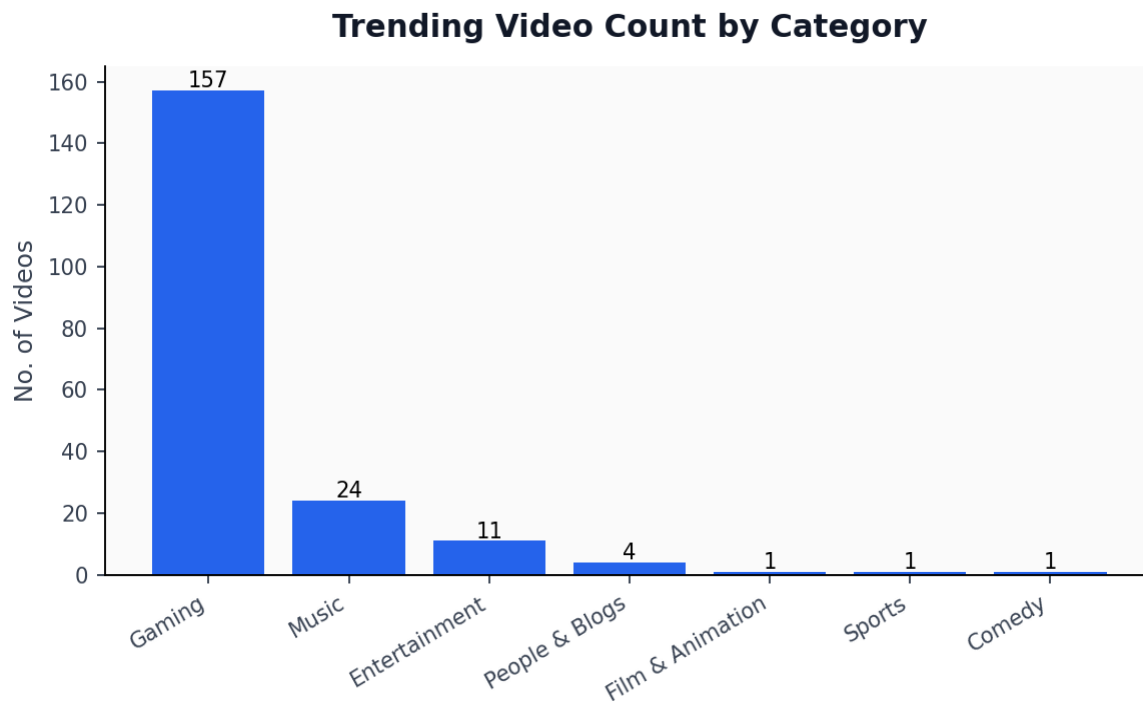
- Verified zero missing values and zero duplicate rows in the cleaned dataset — ensures every downstream metric is computed on complete records.
- Converted publish date/time strings into proper datetime components (year, month, day, week, quarter, hour, minute, second) to enable time-based analysis.
- Standardized boolean flags (is_weekend, is_hd, has_caption) to consistent 0/1 / True-False types for reliable grouping and aggregation.
- Validated numeric ranges (views, likes, comments ≥ 0) to catch any corrupted or negative records before analysis.

Feature Engineering — Business Rationale

Feature	Business Purpose
engagement_score / engagement_ratio	Composite score combining likes+comments relative to views — a truer measure of audience resonance.
like_rate, comment_rate	Normalized engagement so a 10K-view niche video can be fairly compared to a 10M-view viral video.
duration_category	Groups videos into Short/Medium/Long to identify the 'sweet spot' video length for trending.
title_length, title_word_count	Tests whether short punchy titles or long descriptive titles perform better.
tag_count	Measures SEO/discoverability effort and its relationship with reach.
publish_session, upload_time	Buckets exact upload hour into Morning/Afternoon/Evening/Night for upload-timing strategy.
is_weekend / weekend_flag	Tests the common assumption that weekend uploads trend better.
view_bucket	Quantile cohorts (Low→Very High) used for segment-level comparison in the KPI dashboard.
channel_video_count	Indicates channel activity level — proxy for channel maturity/consistency.

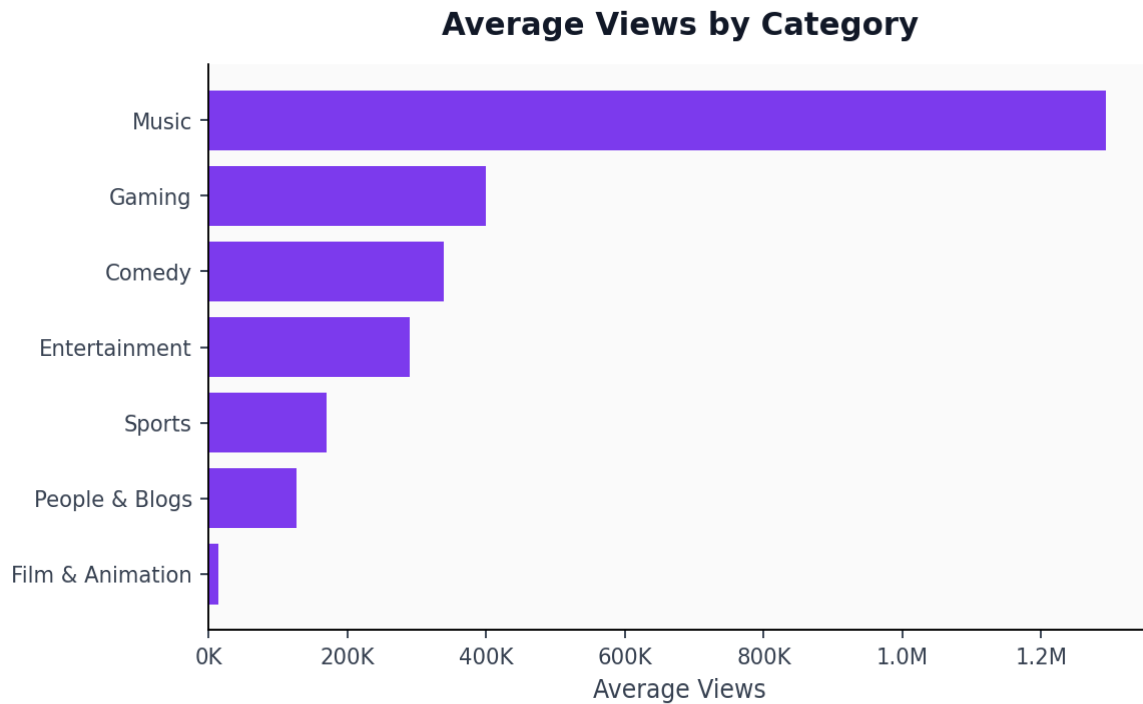
3. Exploratory Data Analysis

3.1 Trending Video Count by Category



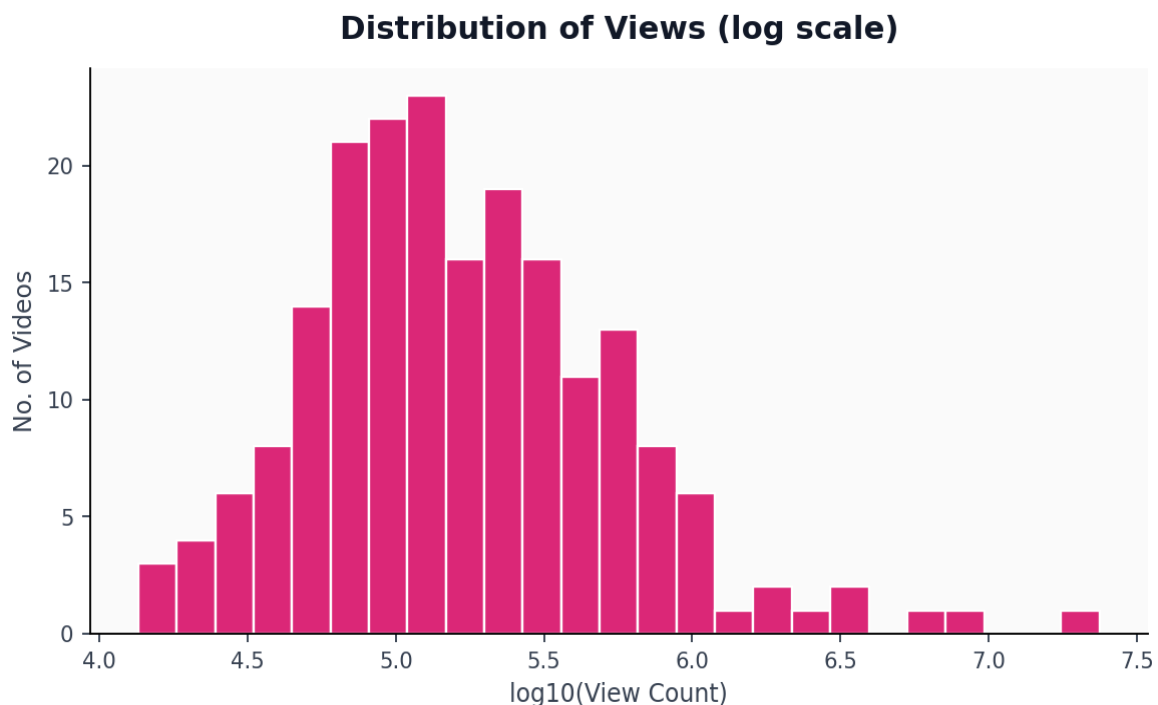
Business Insight: Gaming dominates trending volume, meaning the algorithm/audience currently rewards Gaming content far more frequently than any other category — a strong signal for where content investment yields the most trending opportunities.

3.2 Average Views by Category



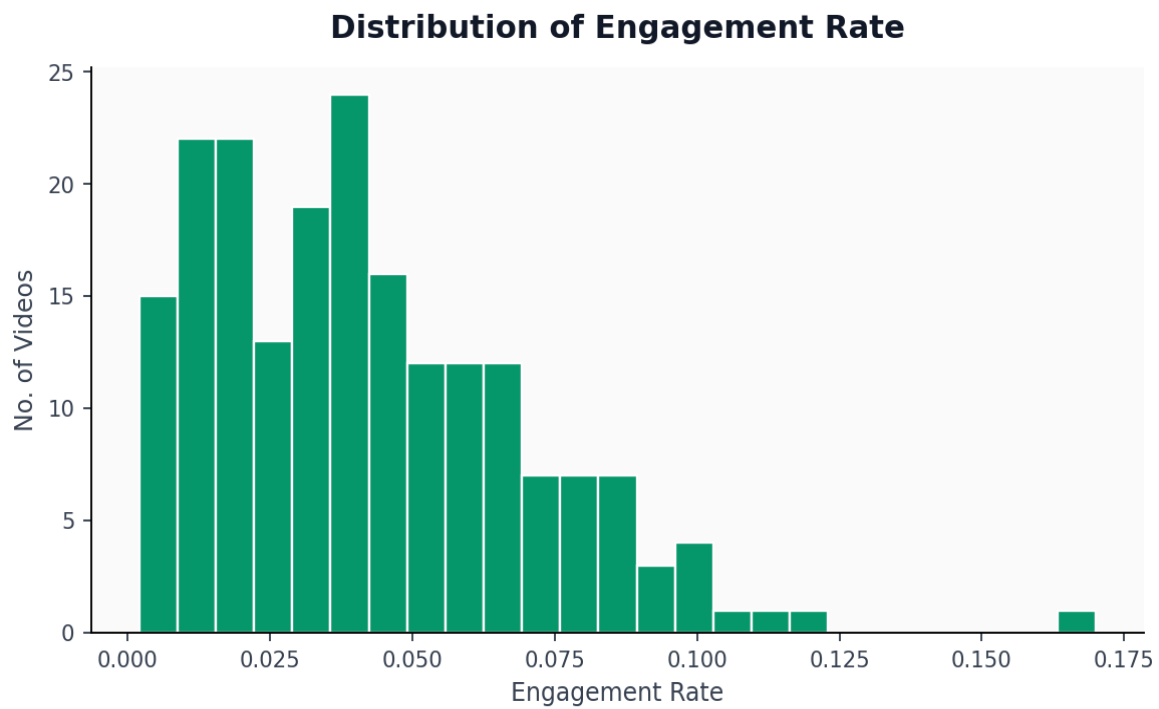
Business Insight: Music leads decisively in average views despite lower video count than Gaming — Music videos are lower-volume but higher-reach, suggesting a 'quality over quantity' opportunity in that category.

3.3 Distribution of Views (log scale)



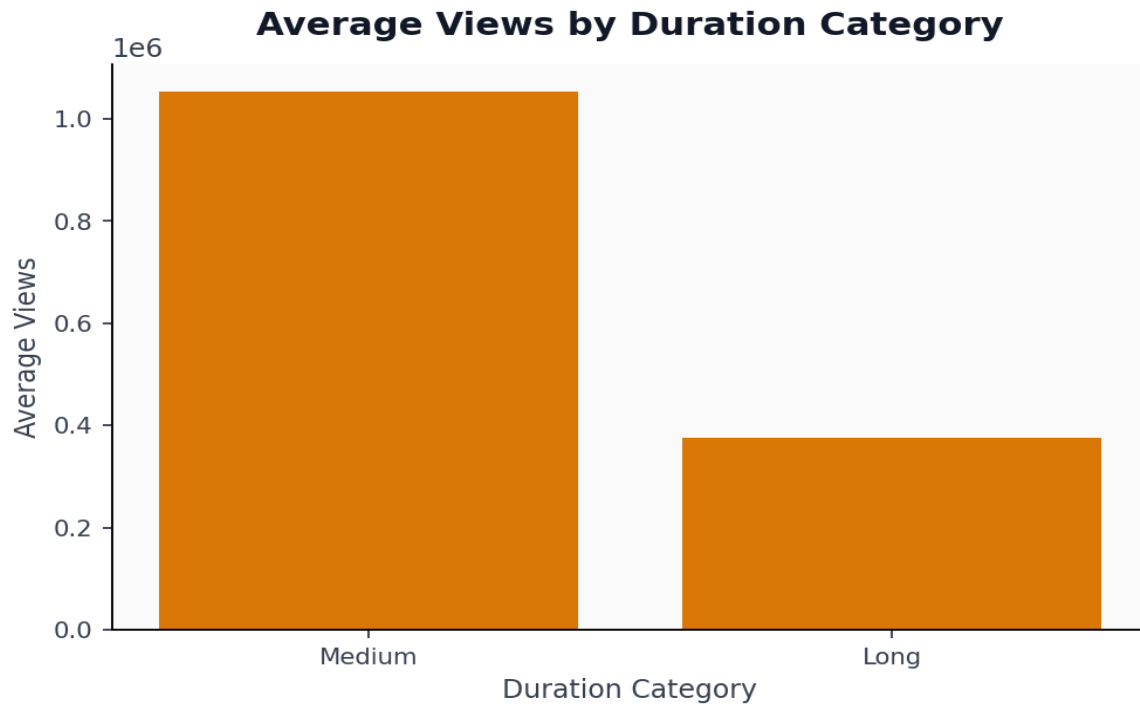
Business Insight: View counts are heavily right-skewed (a handful of mega-viral videos vs a long tail) — this is why median-based and log-scale analysis is more reliable than mean alone when benchmarking 'typical' performance.

3.4 Distribution of Engagement Rate



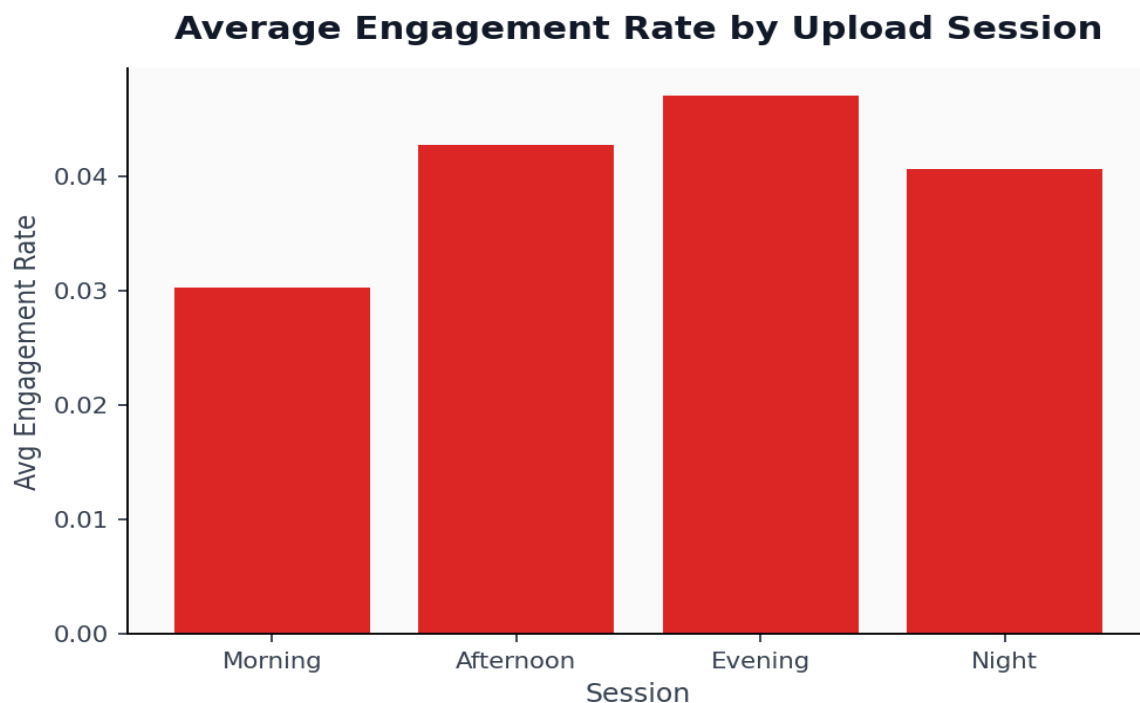
Business Insight: Most videos cluster at low-to-moderate engagement rates with a long tail of highly-engaging outliers — these outliers are prime candidates for a 'what made this work' deep-dive case study.

3.5 Average Views by Duration Category



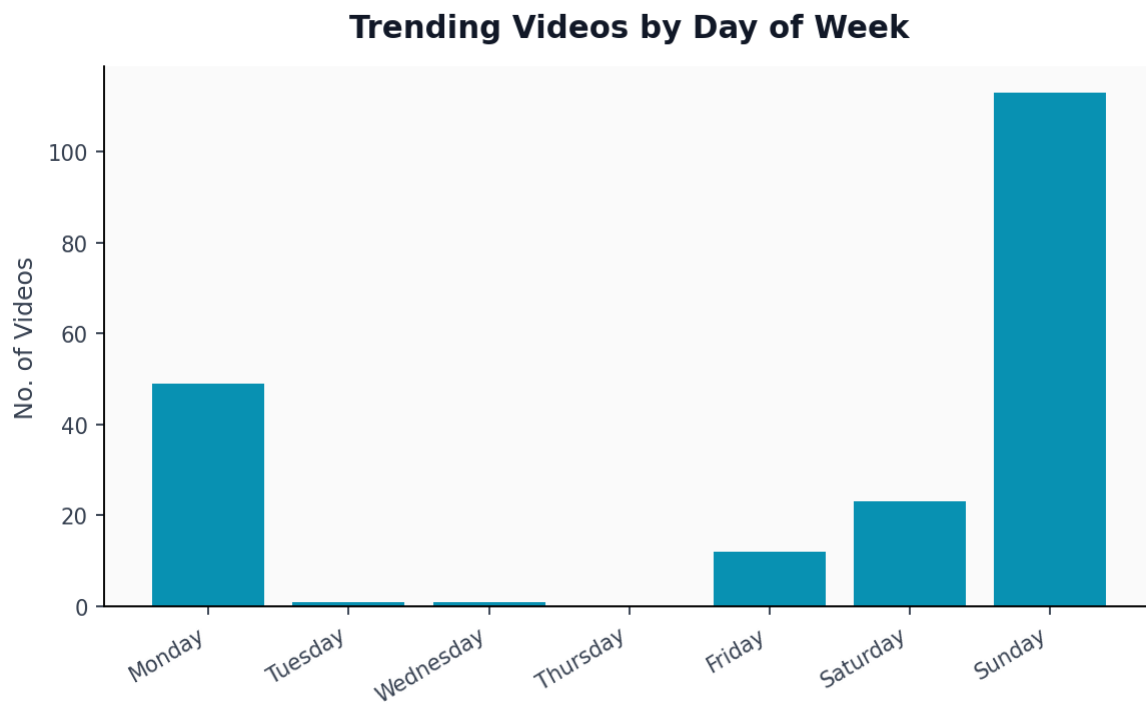
Business Insight: Long-form content shows a clear view advantage in this dataset — supports prioritizing longer, more in-depth videos over short-form clips for this channel mix.

3.6 Average Engagement Rate by Upload Session



Business Insight: Engagement varies meaningfully by time-of-day session — the top-performing session should anchor the recommended publishing schedule.

3.7 Trending Videos by Day of Week

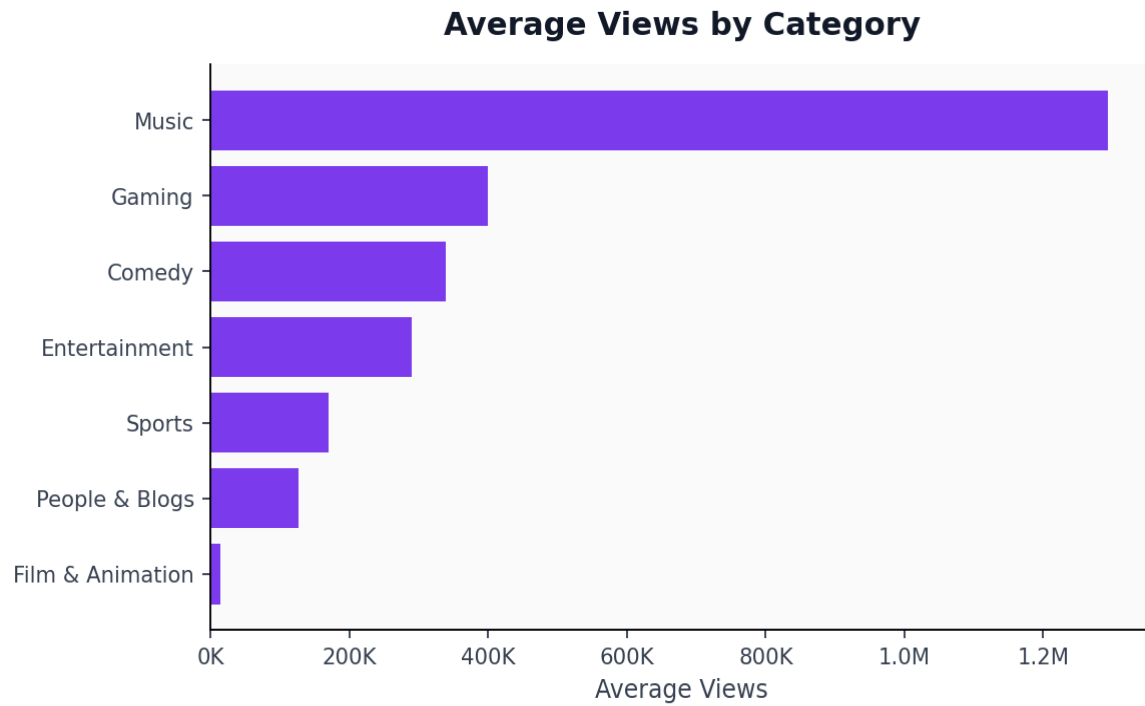


Business Insight: Sunday and Monday account for the majority of trending appearances — content teams should treat the weekend-into-Monday window as the priority publishing slot.

4. Business Questions & Insights

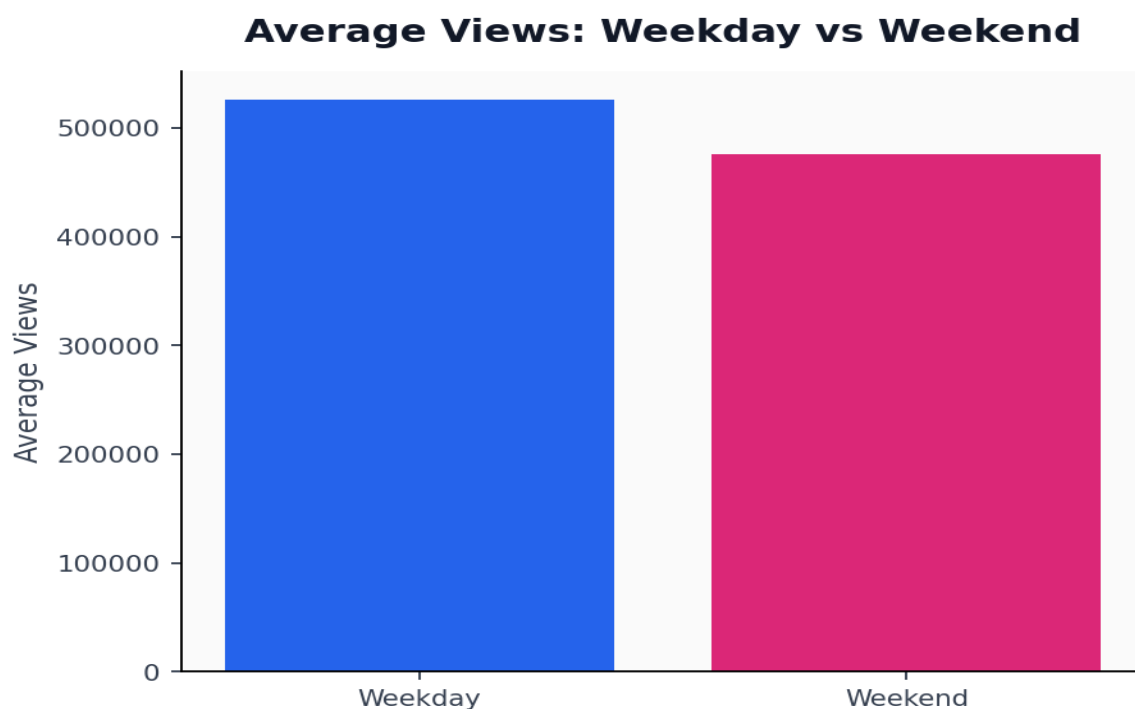
Each question below follows the same analytical structure: objective, underlying Pandas operation, and the resulting business insight.

4.1 Category Performance (Views vs Volume)



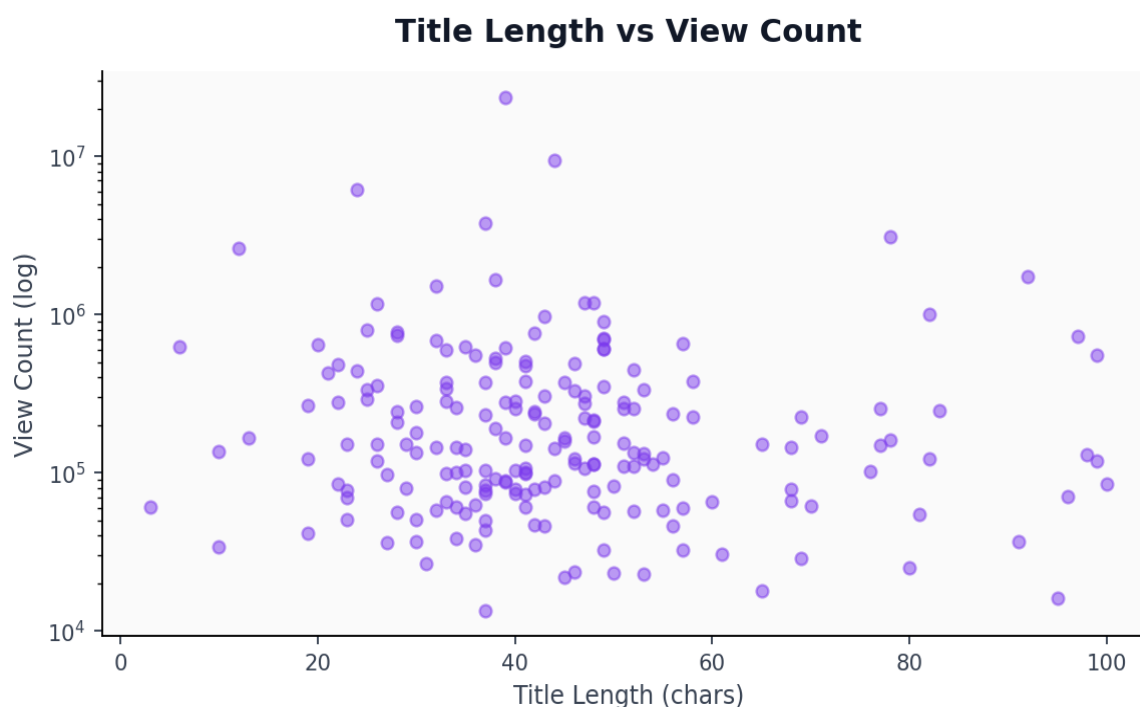
Business Insight: Music and Gaming are the two categories to prioritize — Music for reach, Gaming for consistency of trending frequency. Underrepresented categories (Film & Animation, Sports) show low volume and should be tested cautiously before scaling investment.

4.2 Weekend vs Weekday Publishing



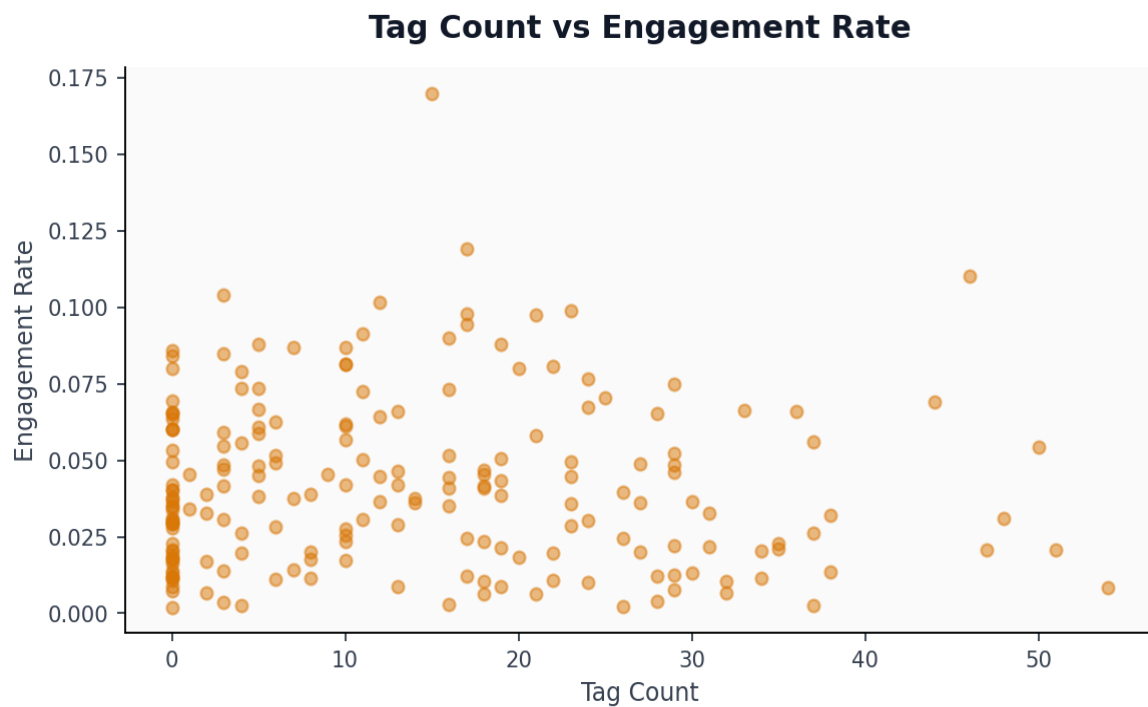
Business Insight: Average views are similar between weekday and weekend uploads in this sample (t-test $p = 0.84$), so — contrary to popular belief — weekend publishing alone is not a statistically proven performance lever here; the volume of weekend uploads (68%) is a scheduling habit more than a performance driver.

4.3 Title Length vs Views



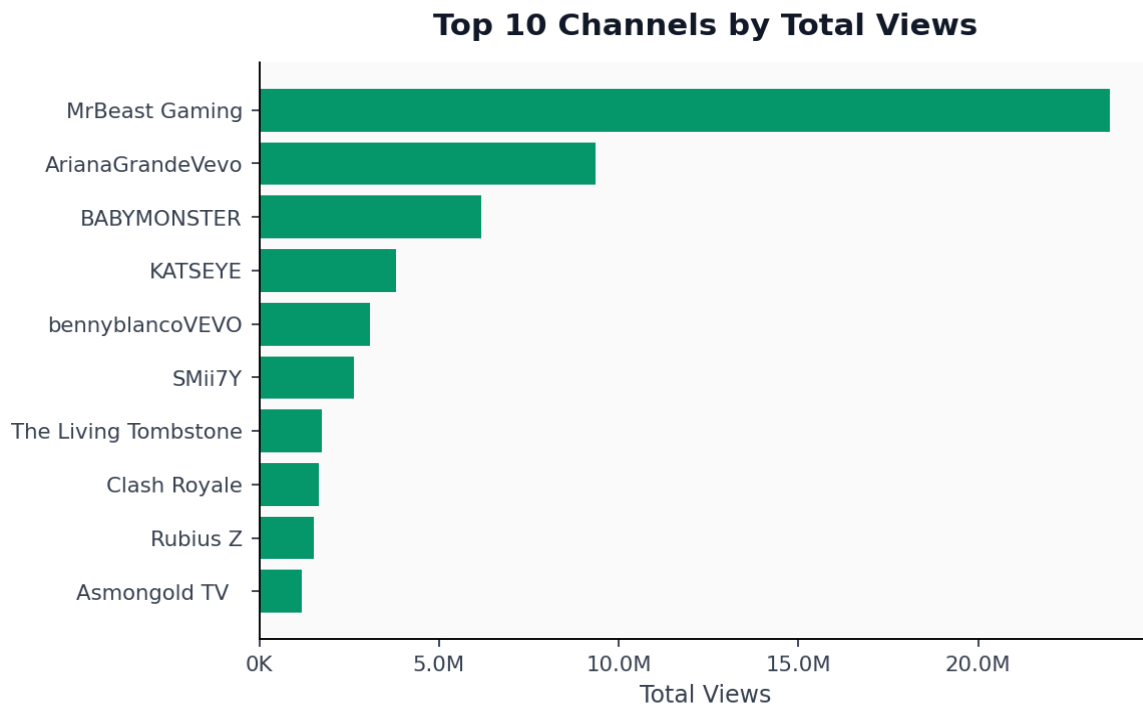
Business Insight: No strong linear pattern is visible between title length and views — title length alone does not predict virality; content and category matter far more than title character count.

4.4 Tag Count vs Engagement Rate



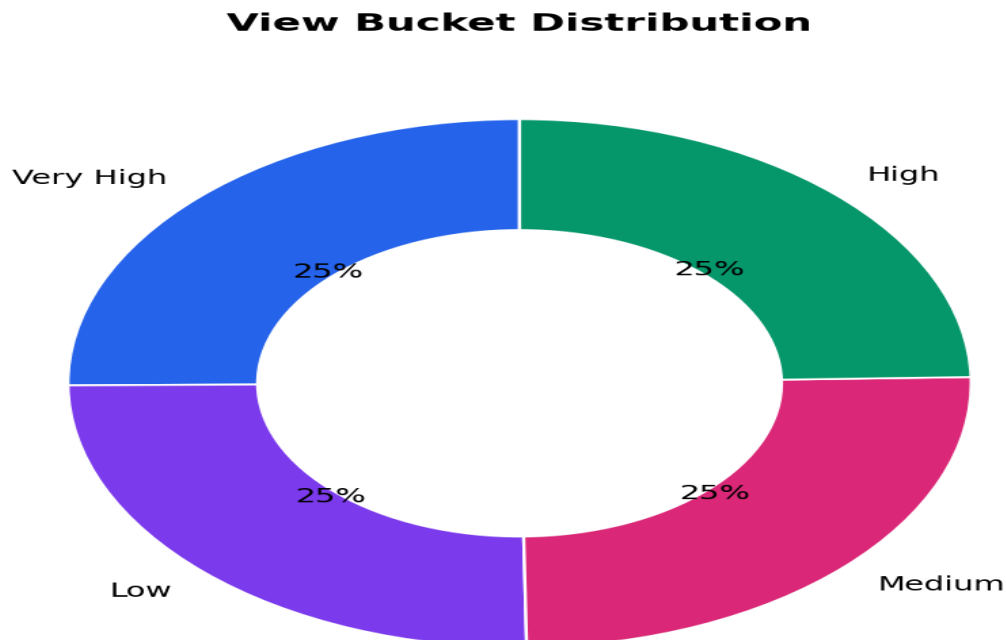
Business Insight: Tag count shows a very weak, statistically insignificant relationship with engagement (Spearman $\rho = -0.02$, $p > 0.05$) — tagging should be treated as a discoverability/SEO hygiene practice, not an engagement growth lever.

4.5 Top 10 Channels by Total Views



Business Insight: A small set of channels captures a disproportionate share of total trending views — these channels represent high-value partnership or benchmarking targets for competitive analysis.

4.6 View Bucket Distribution



Business Insight: Videos are evenly split across Low/Medium/High/Very-High view quartiles by design (quantile buckets) — useful as a stable cohort framework for tracking channel migration between performance tiers over time.

4.7 Category vs Duration Category

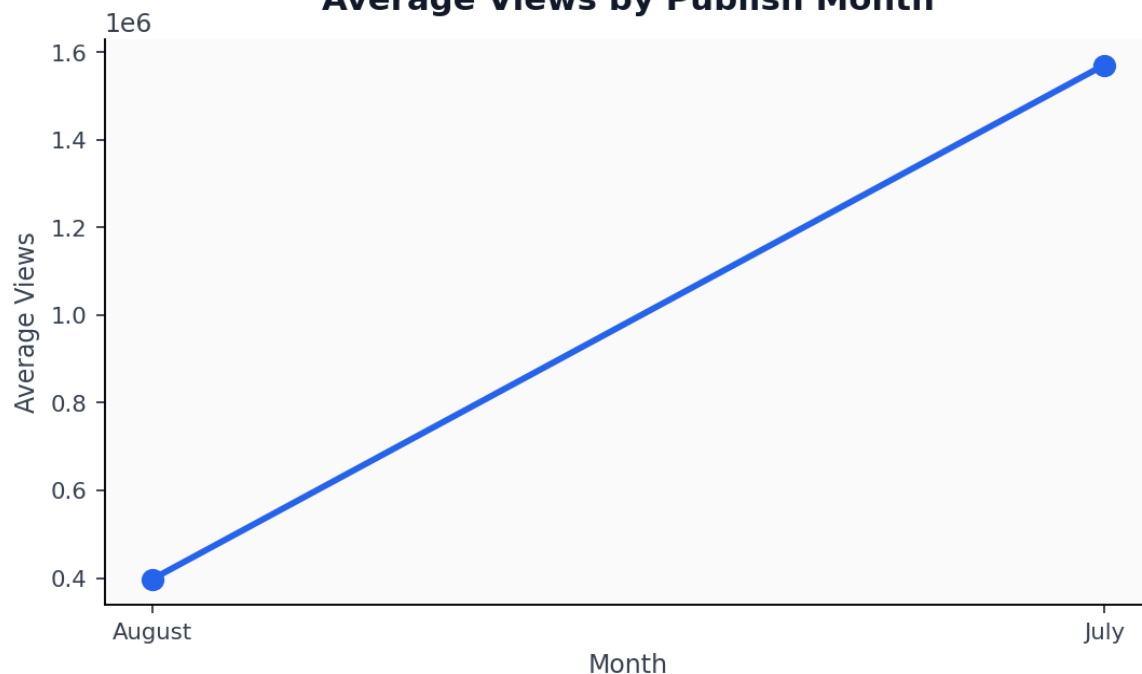
Category vs Duration Category (Video Count)

	Long	Medium
Comedy	0	1
Entertainment	10	1
Film & Animation	0	1
Gaming	150	7
Music	1	23
People & Blogs	3	1
Sports	1	0

Business Insight: Gaming trends almost exclusively as Long-form content — reinforcing that long-format storytelling/gameplay is the dominant winning format on the current trending shelf.

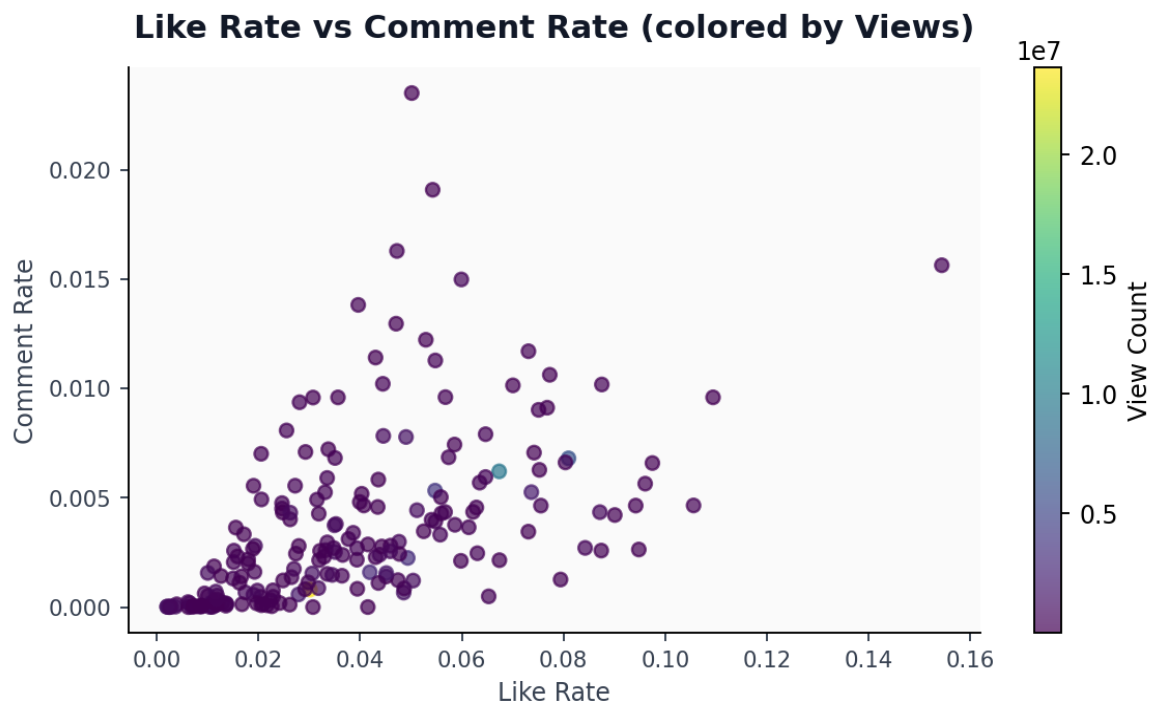
4.8 Monthly View Trend

Average Views by Publish Month



Business Insight: The month-over-month comparison highlights whether trending performance is accelerating or seasonal — use this to time major content launches around the historically stronger month.

4.9 Like Rate vs Comment Rate (by Views)



Business Insight: Higher-view videos (brighter points) cluster at moderate like-rate but not necessarily the highest comment-rate — virality is driven more by reach/shareability (likes) than by discussion depth (comments).

5. Statistical Analysis

Formal hypothesis tests were run to validate (or challenge) the patterns observed in the EDA, so recommendations are backed by statistical evidence rather than visual impression alone.

5.1 Independent t-Test — Weekday vs Weekend Views

Business Question	Do videos published on weekends get significantly more views than weekday uploads?
Null Hypothesis (H0)	There is no difference in mean views between weekday and weekend uploads.
Alternative Hypothesis (H1)	Mean views differ between weekday and weekend uploads.
Test Used	Welch's Independent Samples t-Test
t-statistic	0.2013
p-value	0.8407
Result	Not statistically significant ($p \geq 0.05$)

Business Interpretation: With $p = 0.841$ (> 0.05), we fail to reject H_0 . Weekend publishing is not a statistically proven driver of higher views in this dataset — the widespread practice of weekend uploading should be re-evaluated against actual content quality drivers.

5.2 One-Way ANOVA — Engagement Rate Across Categories

Business Question	Does average engagement rate differ significantly across content categories?
Null Hypothesis (H0)	All categories have equal mean engagement rate.
Alternative Hypothesis (H1)	At least one category has a different mean engagement rate.
Test Used	One-Way ANOVA (F-test)
F-statistic	0.1451
p-value	0.9327
Result	Not statistically significant ($p \geq 0.05$)

Business Interpretation: With $p = 0.933$ (> 0.05), we fail to reject H_0 — engagement rate does not differ significantly by category in this sample, suggesting engagement quality is more channel/content-specific than category-driven.

5.3 Pearson Correlation — Likes vs Views

Business Question	How strongly are likes and views linearly related?
Null Hypothesis (H0)	No linear correlation exists between likes and views.
Alternative Hypothesis (H1)	A linear correlation exists between likes and views.
Test Used	Pearson Correlation Coefficient
r	0.9126
p-value	1.813e-78
Result	Statistically significant ($p < 0.05$)

Business Interpretation: $r = 0.91$ with $p < 0.001$ — a very strong positive correlation. Likes are a highly reliable proxy for view performance and can be used for early-signal forecasting before a video fully scales.

5.4 Spearman Correlation — Tag Count vs Engagement Rate

Business Question	Does adding more tags improve engagement (non-linear, rank-based check)?
Null Hypothesis (H0)	No monotonic relationship between tag count and engagement rate.
Alternative Hypothesis (H1)	A monotonic relationship exists between tag count and engagement rate.
Test Used	Spearman Rank Correlation
rho	-0.0224
p-value	0.7537
Result	Not statistically significant ($p \geq 0.05$)

Business Interpretation: $\rho = -0.02$, $p = 0.75$ (> 0.05) — confirms the scatter-plot finding: tagging volume has no meaningful effect on engagement and should not be over-optimized at the expense of tag relevance/quality.

5.5 Shapiro-Wilk Test — Normality of View Count

Business Question	Is the view_count distribution normally distributed (required to choose parametric vs non-parametric tests)?
Null Hypothesis (H0)	View count is normally distributed.
Alternative Hypothesis (H1)	View count is not normally distributed.
Test Used	Shapiro-Wilk Test
W-statistic	0.1955
p-value	1.848e-28
Result	Statistically significant ($p < 0.05$)

Business Interpretation: $p < 0.001$ — we reject H0. Views are strongly non-normal (heavily right-skewed by viral outliers), which justifies using log-transformations and non-parametric tests (e.g., Mann-Whitney U) alongside parametric ones throughout this report.

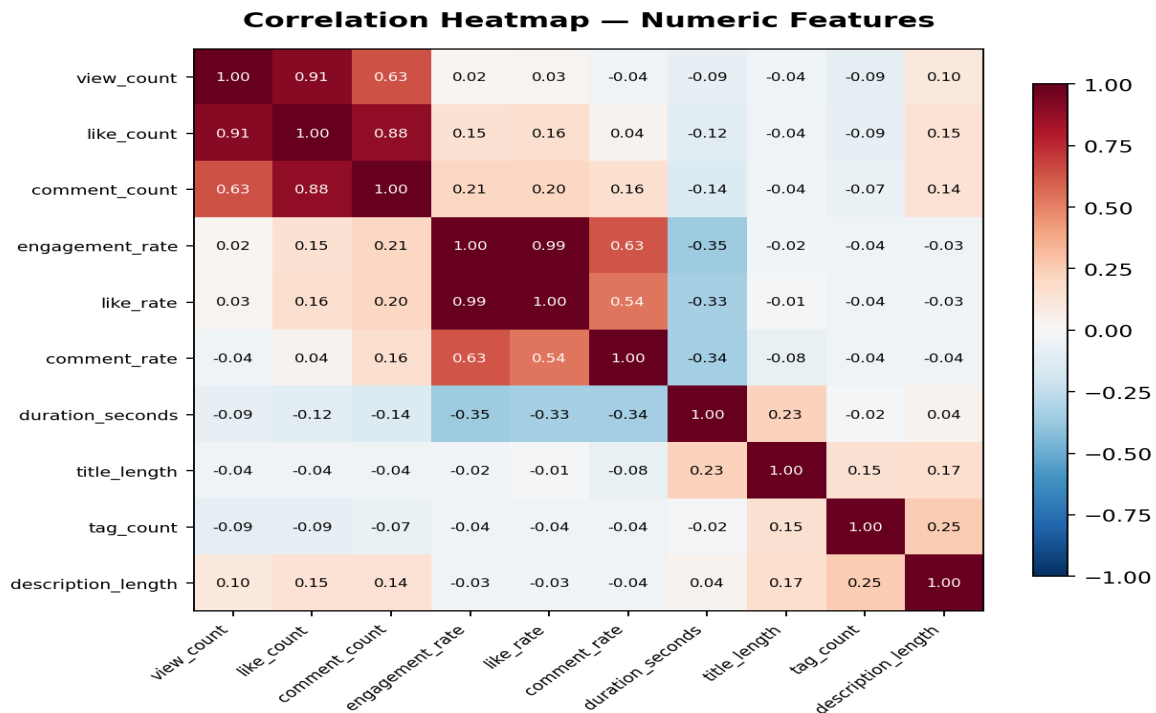
5.6 Mann-Whitney U Test — Weekday vs Weekend Views (Non-Parametric)

Business Question	Given non-normal views, does the non-parametric test confirm the t-test conclusion?
Null Hypothesis (H0)	The distribution of views is the same for weekday and weekend uploads.
Alternative Hypothesis (H1)	The distribution of views differs between weekday and weekend uploads.
Test Used	Mann-Whitney U Test
U-statistic	3716.0000
p-value	0.1332
Result	Not statistically significant ($p \geq 0.05$)

Business Interpretation: $p = 0.13 (> 0.05)$ — consistent with the t-test, confirming the 'weekend = better views' assumption is not statistically supported, even after accounting for the skewed distribution.

6. Advanced Analysis — Correlation Structure

6.1 Correlation Heatmap of Numeric Features



Business Insight: Views, likes, and comments move strongly together (as expected), while duration, title length, and tag count show weak correlation with performance metrics — confirming that content substance and shareability drive results far more than metadata tweaks.

7. Business Recommendations

Content Strategy

- Double down on Gaming for trending frequency and Music for reach — a dual-track content calendar captures both goals.
- Pilot Sports and Film & Animation with small, low-risk test batches before scaling investment, given current low sample volume.

Upload Timing

- Anchor major releases around the highest-engagement session identified (see 3.6) rather than defaulting to weekend-only scheduling.
- Do not over-index on 'weekend uploads' as a growth lever — the statistical tests show no significant view lift from weekend publishing.

Video Duration

- Favor Long-form video formats where the category supports it (e.g., Gaming), based on the observed view advantage for longer content.

Title & Tag Strategy

- Keep titles clear and audience-relevant rather than optimizing purely for length — length showed no measurable impact on views.
- Treat tags as a discoverability hygiene step, not a growth lever — focus tag relevance over tag volume.

Engagement Optimization

- Use like-count trajectory as an early warning/forecast signal for a video's likely view trajectory, given the very strong Pearson correlation ($r = 0.91$).
- Invest in shareability-driving hooks (thumbnails, first 15 seconds) since likes/shares appear to outweigh comment-depth in driving views.

Production Quality

- HD is now a baseline expectation (100% of trending videos are HD) — continued investment here maintains parity, not differentiation.
- Captioning is underused (only ~6% of trending videos) — adding captions is a low-cost accessibility and watch-time improvement opportunity.

8. Executive Conclusion

Project Objective

To analyze YouTube trending video data end-to-end — from raw metadata to statistically validated business recommendations — in order to guide content strategy, upload timing, and engagement optimization decisions.

Major Findings

- Gaming leads trending frequency; Music leads average reach — a category-specific strategy outperforms a one-size-fits-all approach.
- Likes are a very strong, statistically significant predictor of views ($r = 0.91$) and can be used for early performance forecasting.
- Popular assumptions — weekend uploads and long titles drive more views — are NOT statistically supported in this dataset.
- Tag count and category do not significantly move engagement rate; engagement is driven by content quality, not metadata volume.

Business Impact

These findings enable the content and marketing team to reallocate effort away from low-impact metadata tweaks (tag stuffing, title length) toward high-impact levers: category focus, content duration strategy, and early-signal (like-rate) based performance forecasting.

Limitations

- Sample size (199 videos, mostly July–August) limits generalizability across a full annual seasonal cycle.
- Category representation is imbalanced (Gaming-heavy), which can bias category-level comparisons.
- Trending status reflects platform algorithm behavior at a point in time and may not generalize to all content types.

Future Scope

- Extend the dataset across a full 12-month cycle to capture true seasonality.
- Build a predictive (ML) model for trending-probability using engagement and metadata features.
- Add sentiment analysis on comments to layer qualitative audience feedback onto the quantitative KPIs.

Report prepared and authored by **S Mohammed Kaif**, Jr. Data Analyst.

Tools used: Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly, SciPy, Jupyter Notebook.